

Variables

13 June 2026 10:25

What is variable ?

In real life we store things in containers

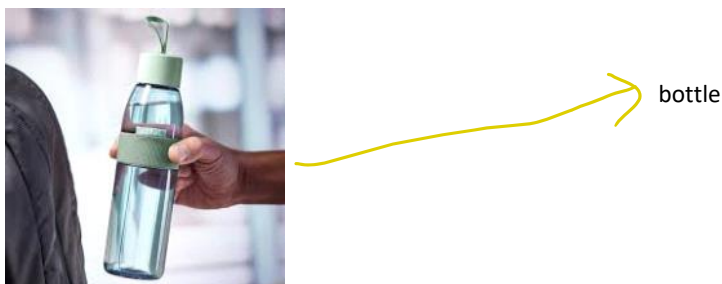
Eg:

1. Water bottle stores water
2. School bag store books
3. Wallet store money

Similarly in java a variable is also like container

Definition

Variable = a named box/container used to store values in java



```
int num=20;
```

Int means the type of value is an integer value

num : is the name of variable in java. This name is also called an identifier.

Why we need variables ?

Without variables

```
System.out.println(20);
System.out.println(20);
System.out.println(20);
```

If num change 20 to 21 we must change it everywhere.

With variable

```
-----
int num=20;
    System.out.println(num);
    System.out.println(num);
    System.out.println(num);
```

Now if num change from 20 to 21, we only change at one place

Basic syntax of variable

dataType variablename = value

Eg:

Int marks=85;

Int --> data type

Marks --> variable name

= --> operator

85 --> value

; ---> statement end

Q1: write java program to print age 20:

Variable declaration and initialization

```
int age=20;
```

Q2: Write java program to take one age variable and print it after it change value and print it

Note: variable name (identifiers) have some rules

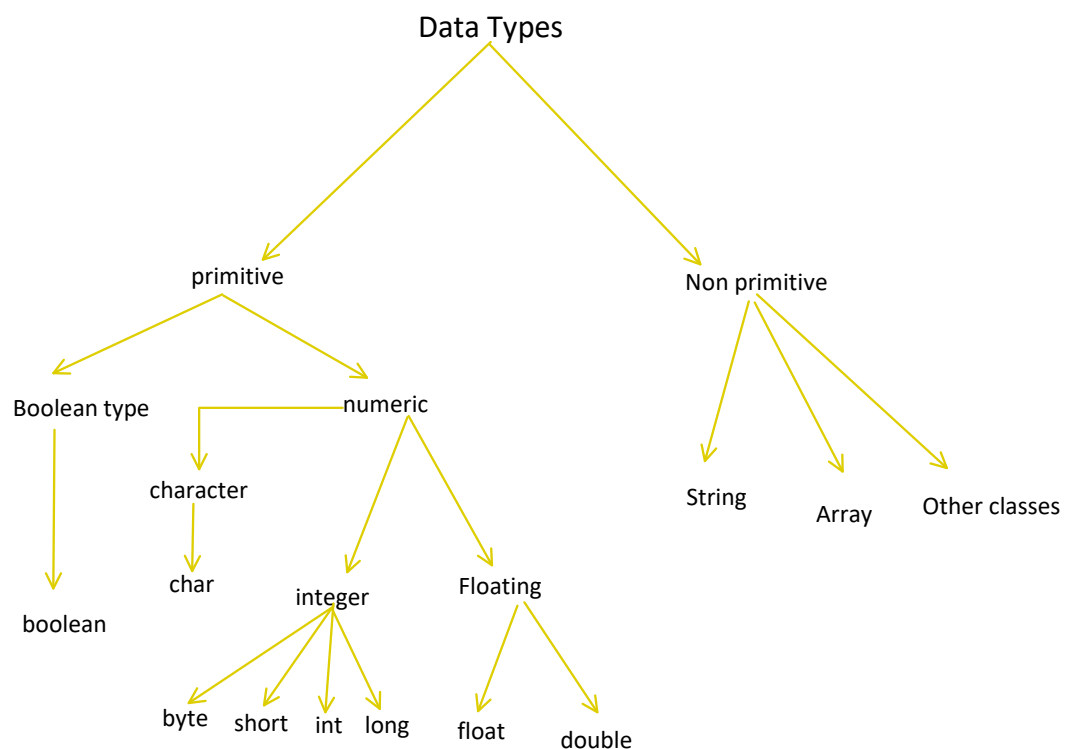
Rules:

1. Lower case, Upper case, digit, \$, _
2. Start with : letter, _ , \$. It cannot start with digit
3. Case sensitive: age, Age, AGE AgE
4. Reserved keywords : int , for, if cannot used

Examples:

1. Age :
2. _marks
3. \$value
4. 1cityName:
5. cityName:
6. #myname:

Break : 11:20



Before understanding data types, first understand data

Name: rahul
Age: 20
Marks : 50
Price: 77.80
Grade : A
Payment done: true

What is Data type ?

Data type = type/category of data

Int age=20

20=20 interger type ki value

age = var name

Variable is the box

Data type tells what kind of item we can put inside the box

Java has 8 primitive data types:

Data type	Used for	Example
byte	Small whole number	100
Short	Whole	30000
Int	Common whole num	23
Long	Very large whole num	9989788686677
Float	Decimal num	10.5f
Double	Common decimal num	99.99
Char	Single char	'A'
Boolean	True/false	true

Data Types	Default Size	Default Value	Comments
byte	1 Byte	0	Best suited for Stream I/O operation
short	2 Bytes	0	Rarely used nowadays
int	4 Bytes	0	It is the default data type of java to hold integer values
long	8 Bytes	0L	It is recommended for the files with size in GB's
float	4 Bytes	0.0f	For 5 point precession
double	8 Bytes	0.0D	For 15 point precession
char	2 Bytes	'\u0000'	It is of 2 Bytes because it supports I18N
boolean	1 bit	false	It store the literal true or false only

8 GB ----> MB -- 1024--> 1KB--->Byte--->bits

8 bit=1 Byte